

Two Explicit Witness Sectors of the Random-Embedding Conjecture

Automated mathematical discovery run `fa_banach_001`
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July 22, 2026

Status. Partial result likely valid; human review needed. The full existential- embedding conjecture remains open.

1 Source conjecture

Hayes, Jekel, and Kunawalkam Elayavalli formulate two conjectures about random embeddings of free group factors [1]. Their stronger Conjecture 1.2, restated as Conjecture 5.1, is the following. Let $U_1^{(N)}, \dots, U_r^{(N)}$ be independent Haar random unitaries, and let $u_1, \dots, u_r$ be the canonical free Haar unitaries in $L(\mathbb{F}_r)$. For every self-adjoint noncommutative $*$ -polynomial p in $r + s$ unitary variables, they predict

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\sup_{V_1, \dots, V_s \in \mathcal{U}(M_N(\mathbb{C}))} \text{tr}_N p(U^{(N)}, V) \right] = \sup_{v_1, \dots, v_s \in \mathcal{U}(L(\mathbb{F}_r))} \tau p(u, v). \quad (1)$$

They prove that this is equivalent to existentiality of the random embedding of $L(\mathbb{F}_r)$ into a matrix ultraproduct.

because Ozawa showed that the AO property implies solidity [Oza02], while our random AO property would imply the Peterson-Thom property which is can be viewed as an optimal strengthening of solidity.

Motivated by the above connection to the AO property, we are able to formulate the following stronger conjecture about random unitaries, involving not only traces of polynomials in r -variables, but traces of self-adjoint polynomials in $r + s$ variables, where we optimize over the s auxiliary variables before substituting the random unitaries in the first r variables.

Conjecture 1.2. *Let $U_1^{(N)}, \dots, U_r^{(N)}$ be independent Haar random unitaries. Then for every $s \in \mathbb{N}$ and every self-adjoint $*$ -polynomial in $r + s$ variables, we have*

$$(1.1) \quad \lim_{N \rightarrow \infty} \mathbb{E} \left[\sup_{V_1, \dots, V_s \in \mathcal{U}(M_N(\mathbb{C}))} \text{tr}_N (p(U_1^{(N)}, \dots, U_r^{(N)}, V_1, \dots, V_s)) \right] = \sup_{v_1, \dots, v_s \in \mathcal{U}(L(\mathbb{F}_r))} \tau(p(u_1, \dots, u_r, v_1, \dots, v_s)).$$

In Theorem 1.3 (2), we will relate Conjecture 1.2 to the concept of existential embeddings of operator algebras, which comes from continuous model theory [FHS14]. An embedding $Q \leq M$ of tracial von Neumann algebras is *existential* if a trace-preserving embedding $\theta: M \rightarrow Q^\omega$ so that $\theta|_Q$ is the diagonal embedding; here if Q and M are separable, then ω can be any ultrafilter on the natural numbers, but in the non-separable case, one must use an ultrafilter on a larger index set. The relevant

Figure 1: Conjecture 1.2 in the source paper, page 3.

We verify (1) for two infinite witness sectors. The first consists of all trace-affine witness polynomials. The second consists of separated terms in which one witness conjugates one self-adjoint base polynomial.

2 Proof intuition

The witness optimization is difficult when several occurrences of the auxiliary unitaries interact. In the two sectors below, it has an exact spectral solution before taking $N \rightarrow \infty$.

For a trace-affine term, polar decomposition gives

$$\sup_{V \in \mathcal{U}(M_N)} \operatorname{Re} \operatorname{tr}_N(CV) = \operatorname{tr}_N |C|.$$

Thus the optimized formula depends only on the singular-value distribution of a base polynomial. For a single conjugation term, the rearrangement inequality gives

$$\sup_{V \in \mathcal{U}(M_N)} \operatorname{tr}_N(AVBV^*) = \int_0^1 \lambda_A(t) \lambda_B(t) dt,$$

where λ_A, λ_B are decreasing eigenvalue curves. The same two identities hold in the diffuse finite factor $L(\mathbb{F}_r)$. Almost-sure *-moment convergence of Haar matrices then gives convergence of trace norms and of the decreasing eigenvalue curves. This proves not only the expected limit in the conjecture, but an almost-sure limit for every fixed polynomial in either sector.

3 Exact optimization lemmas

Throughout, a finite factor carries its normalized trace τ . For a self-adjoint A in such a factor, let $\lambda_A : [0, 1] \rightarrow \mathbb{R}$ denote its decreasing eigenvalue rearrangement, or spectral scale. For a matrix it is the step function formed from the eigenvalues in decreasing order.

Lemma 1 (Polar optimization). *For every C in a finite factor,*

$$\sup_{V \in \mathcal{U}(\mathcal{M})} \operatorname{Re} \tau(CV) = \tau(|C|). \quad (2)$$

Proof. The upper bound follows from noncommutative Hölder inequality. Write $C = W|C|$. The polar partial isometry W^* extends to a unitary V because its initial and final complements have the same trace. For this extension, VW is the support projection of $|C|$, and cyclicity gives $\tau(CV) = \tau(|C|VW) = \tau(|C|)$. $\square$

Lemma 2 (Rearrangement optimization). *If A, B are self-adjoint elements of a diffuse finite factor, then*

$$\sup_{V \in \mathcal{U}(\mathcal{M})} \tau(AVBV^*) = \int_0^1 \lambda_A(t) \lambda_B(t) dt. \quad (3)$$

Proof. The upper bound is the Hardy–Littlewood rearrangement inequality for finite von Neumann algebras; it follows from the generalized singular-value inequalities of Fack and Kosaki [2]. The usual positive-operator form implies the self-adjoint form after adding sufficiently large scalar multiples of the identity to A and B ; all added terms are independent of V .

For completeness, the reverse inequality can be seen directly. Using spectral scales, approximate A and B in L_2 by decreasing step operators

$$A_m = \sum_{k=1}^m a_k p_k, \quad B_m = \sum_{k=1}^m b_k q_k,$$

where the p_k and q_k are orthogonal partitions of the identity with $\tau(p_k) = \tau(q_k) = 1/m$, and a_k, b_k are decreasing. Atoms of a spectral measure may be split inside their spectral projections because the factor is diffuse. Projection families with the same trace vector are unitarily conjugate, so there is V_m with $V_m q_k V_m^* = p_k$ for every k . Therefore

$$\tau(A_m V_m B_m V_m^*) = \frac{1}{m} \sum_{k=1}^m a_k b_k.$$

The right side tends to the integral in (3). The L_2 approximation and Cauchy–Schwarz show that replacing A_m, B_m by A, B changes the trace by $o(1)$, proving the lower bound. $\square$

4 The two witness sectors

Let $q = q^*$ and all other capital letters below denote fixed noncommutative $*$ -polynomials in the base variables $X_1, \dots, X_r$.

Theorem 3. *For every fixed polynomial p of either form below, the optimized random matrix value converges almost surely to the $L(\mathbb{F}_r)$ value, and (1) holds.*

(A) Trace-affine sector. For arbitrary $C_1, \dots, C_s$,

$$p(X, Y) = q(X) + \sum_{j=1}^s (C_j(X) Y_j + Y_j^* C_j(X)^*). \quad (4)$$

This includes, modulo cyclicity of the trace, every self-adjoint polynomial of total witness degree at most one.

(B) Separated one-conjugation sector. For self-adjoint $A_j = A_j^*$ and $B_j = B_j^*$,

$$p(X, Y) = q(X) + \frac{1}{2} \sum_{j=1}^s (A_j(X) Y_j B_j(X) Y_j^* + Y_j B_j(X) Y_j^* A_j(X)). \quad (5)$$

Proof. We use Voiculescu’s almost-sure asymptotic freeness theorem [3], which gives, for every fixed $*$ -polynomial R ,

$$\mathrm{tr}_N R(U^{(N)}) \longrightarrow \tau R(u) \quad \text{almost surely.} \quad (6)$$

All polynomial evaluations under consideration have a common uniform operator-norm bound, since every variable is unitary.

For (A), Lemma 1, applied independently to every witness, gives the exact finite- N formula

$$\sup_V \mathrm{tr}_N p(U^{(N)}, V) = \mathrm{tr}_N q(U^{(N)}) + 2 \sum_{j=1}^s \mathrm{tr}_N |C_j(U^{(N)})|. \quad (7)$$

The identical factor formula is

$$\sup_v \tau p(u, v) = \tau q(u) + 2 \sum_{j=1}^s \tau |C_j(u)|. \quad (8)$$

By (6), the constant terms converge. To treat the trace norms, apply (6) to powers of the positive matrices

$$C_j(U^{(N)})^* C_j(U^{(N)}).$$

Uniform polynomial approximation of $t \mapsto \sqrt{t}$ on a common compact interval yields

$$\mathrm{tr}_N |C_j(U^{(N)})| \longrightarrow \tau |C_j(u)|.$$

Equations (7) and (8) prove the almost-sure assertion for the affine sector.

For (B), cyclicity and Lemma 2 give

$$\sup_V \mathrm{tr}_N p(U^{(N)}, V) = \mathrm{tr}_N q(U^{(N)}) + \sum_{j=1}^s \int_0^1 \lambda_{A_j(U^{(N)})}(t) \lambda_{B_j(U^{(N)})}(t) dt, \quad (9)$$

$$\sup_v \tau p(u, v) = \tau q(u) + \sum_{j=1}^s \int_0^1 \lambda_{A_j(u)}(t) \lambda_{B_j(u)}(t) dt. \quad (10)$$

Moment convergence and the common compact spectral bound imply weak convergence of the empirical spectral measure of every self-adjoint base polynomial to its spectral measure in $L(\mathbb{F}_r)$. Generalized inverses of uniformly compactly supported distribution functions therefore satisfy

$$\lambda_{A_j(U^{(N)})}(t) \longrightarrow \lambda_{A_j(u)}(t) \quad \text{for almost every } t,$$

and likewise for B_j . The eigenvalue curves are uniformly bounded, so dominated convergence gives convergence of every integral in (9) to its counterpart in (10).

Finally, in both sectors the optimized quantity is bounded uniformly in N by a constant depending only on p . Dominated convergence upgrades the almost-sure limit to convergence of expectations, which is precisely (1). $\square$

5 Verification, scope, and novelty check

The included numerical script constructs the polar and eigenvector-alignment optimizers for 180 random matrix pairs of orders 2 through 10. Both exact finite-dimensional formulas pass to tolerance 10^{-10} . This is a smoke test only; the proof above is analytic.

The full conjecture remains open. The theorem does not handle monomials with mixed witnesses, higher witness degree, or simultaneous affine and conjugation terms involving the same witness. Those interactions are not reduced to one-variable spectral data by the present argument.

Cheap run-index searches found no packet or attempt for arXiv:2606.02985. Bounded primary searches on July 22, 2026 used the arXiv id, title, exact Conjecture 1.2 label, “existential random embedding,” and trace-affine unitary optimization. They found the source and standard background on trace rearrangements, but no later resolution or explicit record of these two witness sectors.

6 Human-review recommendation

Review as a likely valid substantial partial result. The affine sector is elementary after the exact polar formula. The highest-value checks are the diffuse-factor lower bound in Lemma 2, and the use of weak spectral convergence to pass to the eigenvalue-rearrangement pairing. An expert should also assess whether these explicit sectors are already regarded as folklore in the model theory/free probability community.

References

- [1] B. Hayes, D. Jekel, and S. Kunnawalkam Elayavalli, *Questions on the structure of random embeddings of $L(\mathbb{F}_2)$* , arXiv:2606.02985v1, 2026.
- [2] T. Fack and H. Kosaki, *Generalized s -numbers of τ -measurable operators*, Pacific J. Math. 123 (1986), no. 2, 269–300.
- [3] D. Voiculescu, *Limit laws for random matrices and free products*, Invent. Math. 104 (1991), no. 1, 201–220.